

12. Implement Iris Flower Classification using KNN

Aim

To implement **Iris Flower Classification using K-Nearest Neighbours**.

Software Required

- Python 3.x
- Scikit-learn
- NumPy

Procedure

1. Load the Iris dataset.
2. Separate features and target.
3. Split the dataset.
4. Create a KNN classifier.
5. Train the classifier.
6. Predict the flower species.
7. Calculate accuracy.

Code

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score

iris = load_iris()
X = iris.data
y = iris.target
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
model = KNeighborsClassifier(n_neighbors=3)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
```

```
print("Predicted Classes:")
print(y_pred)
print("\nAccuracy:",
      accuracy_score(y_test, y_pred))
sample = [[5.1, 3.5, 1.4, 0.2]]
print("\nSample Prediction:",
      iris.target_names[model.predict(sample)[0]])
```

Output:

```
Predicted Classes:
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0]
```

```
Accuracy: 1.0
```

```
Sample Prediction: setosa
```

Sample Prediction: setosa

Result

Thus, Iris flower classification was successfully performed using the **KNN algorithm**.